CHARACTERIZATION OF CONSTRUCTIBLE SUBSETS OF A NOETHERIAN SPACE BY DENSITY CONDITIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that a constructible subset of a Noetherian space is a finite union of locally closed subsets. Invoked inside the counted proof (source lines 3164-3170 and 3187-3192). Turns constructibility into a finite union of locally closed pieces.
- The decomposition of a Noetherian space into finitely many irreducible components. Invoked inside the counted proof (source lines 3176-3178). Splits the minimal counterexample into components.

Source reference. Original lemma lemma-Noetherian: sources/stacks-topology/pinned-original-source.tex, source lines 1470–1470. Navigation pointer only: the counted text refers to the Stacks item lemma-Noetherian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-constructible-Noetherian-space: sources/stacks-topology/pinned-original-source.tex, source lines 3088–3088. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-Noetherian-space, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-constructible-map-Noetherian: sources/stacks-topology/pinned-original-source.tex, source lines 3100–3100. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-map-Noetherian, which is an existing labeled result of this same chapter and is not part of this card.

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Lemma 0.1. *Let X be a Noetherian topological space. Let $E \subset X$ be a subset. The following are equivalent:*

(1) *E is constructible in X, and*
(2) *for every irreducible closed $Z \subset X$ the intersection $E \cap Z$ either contains a nonempty open of Z or is not dense in Z.*

Proof. Assume E is constructible and $Z \subset X$ irreducible closed. Then $E \cap Z$ is constructible in Z by Lemma lemma-constructible-map-Noetherian. Hence $E \cap Z$ is a finite union of nonempty locally closed subsets T_i of Z. Clearly if none of the T_i is open in Z, then $E \cap Z$ is not dense in Z. In this way we see that (1) implies (2).

Conversely, assume (2) holds. Consider the set $\mathcal{S}$ of closed subsets Y of X such that $E \cap Y$ is not constructible in Y. If $\mathcal{S} \neq \emptyset$, then it has a smallest element Y as X is Noetherian. Let $Y = Y_1 \cup \ldots \cup Y_r$ be the decomposition of Y into its irreducible components, see Lemma lemma-Noetherian. If $r > 1$, then each $Y_i \cap E$ is constructible in Y_i and hence a finite union of locally closed subsets of Y_i. Thus $E \cap Y$ is a finite union of locally closed subsets of Y too and we conclude that $E \cap Y$ is constructible in Y by Lemma lemma-constructible-Noetherian-space. This is a contradiction and so $r = 1$. If $r = 1$, then Y is irreducible, and by assumption (2) we see that $E \cap Y$ either (a) contains an open V of Y or (b) is not dense in Y. In case (a) we see, by minimality of Y, that $E \cap (Y \setminus V)$ is a finite union of locally closed subsets of $Y \setminus V$. Thus $E \cap Y$ is a finite union of locally closed subsets of Y and is constructible by Lemma lemma-constructible-Noetherian-space. This is a contradiction and so we must be in case (b). In case (b) we see that $E \cap Y = E \cap Y'$ for some proper closed subset $Y' \subset Y$. By minimality of Y we see that $E \cap Y'$ is a finite union of locally closed subsets of Y' and we see that $E \cap Y' = E \cap Y$ is a finite union of locally closed subsets of Y and is constructible by Lemma lemma-constructible-Noetherian-space. This contradiction finishes the proof of the lemma. □